Fiscal Accountability or Regime Shift? Machine Learning and Causal Evidence from Philippine Gubernatorial Elections (1992–2022)

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***Abstract*—Do voters reward governors for better health and education spending, increased local revenue, and lower dependence on intergovernmental transfers? Using a panel of Philippine provinces (1992–2022) and an XGBoost classifier, we find that while the model achieves 95.5% cross-validated accuracy within the training period, a rolling-window temporal validation reveals catastrophic generalisation failure: accuracy declines to 37–57% for all out-of-sample election years, and to 49.4% for the 2019–2022 period. The central finding is not predictive success but temporal instability. The high within-sample accuracy is an artefact of random splitting; fiscal–electoral relationships are not stable over time. The structural break coincides with the 2016 national political transition, which fundamentally reshaped local accountability dynamics. SHAP analysis identifies the interaction between dynasty status and IRA dependence as the most important predictor, followed by public welfare spending growth—a pattern consistent with clientelistic rather than programmatic accountability. Regression discontinuity analysis estimates that winning a close election increases subsequent health spending growth (ATE = 1.50, 95% CI [0.07, 2.94]), which, together with the predictive model results, suggests a *virtuous cycle*: health spending predicts re-election, and incumbency enables further health investment. We conclude that predictive models of electoral accountability require temporal validation as a minimum standard of rigour, and that the post-2016 period constitutes a regime shift in Philippine local politics with implications for subnational governance and democratic accountability.**

***Index Terms*—fiscal performance, gubernatorial elections, XGBoost, regression discontinuity, temporal validation, political dynasties, Philippines**

I. Introduction

The 1991 Local Government Code substantially devolved fiscal and administrative responsibilities to Philippine local governments, endowing governors with meaningful authority over health, education, and infrastructure budgets. Under canonical models of democratic accountability, such devolution should strengthen electoral incentives: voters observe local service delivery and reward or punish incumbents at the ballot box accordingly. Yet the empirical record on whether fiscal performance actually determines gubernatorial re-election remains limited, particularly with respect to modern machine learning methods capable of capturing nonlinear and interactive effects.

A critical but frequently neglected question concerns *temporal stability*: do the fiscal–electoral relationships identified in one period generalise to future elections? This paper demonstrates that they do not. Using an XGBoost model trained on a 30-year provincial panel, we show that high within-sample cross-validated accuracy (95.5%) is a methodological artefact of random data splitting rather than evidence of a stable predictive relationship. When subjected to rolling-window temporal validation—the appropriate standard for non-stationary political data—model performance is barely distinguishable from random guessing. The collapse occurs sharply after 2016, the year of Rodrigo Duterte's presidential election, which appears to have fundamentally altered local political incentives. Documenting this temporal generalisation failure, and its implications for the methodology of electoral accountability research, constitutes the paper's primary contribution.

II. Theoretical Framework

Two principal theoretical frameworks motivate the empirical analysis. First, *retrospective voting theory* holds that voters evaluate incumbent performance based on observable outcomes—particularly service delivery and economic management—and sanction or reward incumbents accordingly at the ballot box [1]. On this account, governors who expand health and education provision should enjoy higher re-election probabilities, since such investments are visible and attributable. Second, *clientelism theory* posits that politicians in weakly institutionalised democracies secure electoral support through targeted, divisible benefits rather than broad public goods [2]. Rather than investing in programmatic service delivery, clientelistic incumbents use welfare transfers and patronage networks to build loyal electoral coalitions. The distinction between these two mechanisms is empirically testable: retrospective voting predicts that public goods spending (health, education) drives re-election, whereas clientelism predicts that targetable transfers (public welfare, patronage) are more electorally decisive.

III. Data and Variables

A. Data Sources

The analysis draws on two primary datasets. First, a fiscal panel covering 227 local government units (LGUs) from 1992 to 2022, containing annual expenditures on health, education, and public welfare, together with local revenue collections, internal revenue allotment (IRA) share, and the effective number of candidates (ENC). A dynasty indicator was constructed by comparing the surnames of successive incumbents within each LGU. Second, gubernatorial election results for the same period, including vote shares, winners, and turnout. Population figures from five Philippine Statistics Authority censuses (1995–2024) were interpolated linearly to produce annual estimates for per-capita expenditure calculations. Both fiscal and election data were sourced from the Philippine Local Government Interactive Dataset [3].

B. Data Construction

For each LGU–election year combination, we identify the start of an incumbent's term, compute pre-term spending averages (three years prior to the election), and merge fiscal records with electoral outcomes. Per-capita spending is calculated by dividing nominal expenditures (converted from millions to pesos) by interpolated population. Expenditure growth rates are computed as $(\text{post} - \text{pre})/\text{pre}$ and winsorised to $[-0.9, 5]$ to mitigate the influence of extreme outliers. The analytic sample is restricted to terms in which the incumbent sought re-election—defined as the winner of term t appearing as a candidate in term $t + 3$. The final dataset comprises 1,682 observations, of which 787 represent successful re-election bids and 895 represent defeats.

C. Variables

Table I presents the full feature set and the binary target variable. Income class was excluded from the model because all 1,682 observations belong to income class 3 (middle-income), rendering the variable analytically uninformative.

Note on notation: Throughout the text, the Greek letter Δ denotes proportional growth. In computational outputs (e.g., Figure 4), the same variables appear with the prefix `delta_` (e.g., `delta_pubwelf`, `delta_health`). The two forms refer to identical quantities.

TABLE I: Features and Target Variable

Feature	Description
Δhealth	Growth in health spending per capita
Δeduc	Growth in education spending per capita
Δpubwelf	Growth in public welfare spending per capita
Δlocal_rev	Growth in local revenue per capita
ira_share	IRA as a share of total income (higher = more dependent)
enc_go	Effective number of candidates
dynasty	1 if incumbent shares surname with previous governor
prev_margin	Incumbent's vote share in the previous election
region	17 regional dummies (one-hot encoded)
Target: won	1 if re-elected; 0 otherwise

IV. Methodology

A. Predictive Model

We employ an XGBoost classifier [5] with 200 estimators, maximum tree depth of 5, a learning rate of 0.05, and subsample and column-sample ratios of 0.8. Predictive performance is assessed via 5-fold stratified cross-validation, which ensures that the class distribution of the target is preserved across folds. Feature importance is quantified using SHAP (SHapley Additive exPlanations) values [4], which provide theoretically grounded, model-agnostic attributions that are not subject to the high-cardinality bias known to affect XGBoost's native gain-based importance scores. To capture heterogeneous effects of fiscal variables by dynasty status, four interaction terms are added: dynasty × Δeduc, dynasty × Δhealth, dynasty × Δpubwelf, and dynasty × ira_share. Native gain-based importance rankings were computed for completeness but are omitted from the main text for brevity (available upon request); the ordinal rankings are broadly consistent with the SHAP-based results.

B. Temporal Validation

To guard against look-ahead bias and to assess whether identified fiscal–electoral patterns are stable over time, we implement a rolling-window validation protocol. For each election year in the panel, the model is trained exclusively on all preceding election cycles and evaluated on the focal year alone. This design replicates the conditions of real-world electoral forecasting and is the appropriate evaluation standard when the underlying data-generating process may be non-stationary. We additionally compare predictive performance across the pre-2016 and post-2016 sub-periods to assess whether a structural break is present.

C. Regression Discontinuity Design

To move beyond correlation toward causal inference, we exploit the discontinuous assignment of electoral victory in close gubernatorial races. The running variable is the incumbent's vote margin, defined as vote share minus 0.5, such that positive values indicate victory. The treatment is incumbency (margin ≥ 0). Outcomes of interest are subsequent health spending growth and post-election IRA share. Local linear regression is estimated with a triangular kernel at bandwidths of 3%, 5%, and 7%, following the recommendations of Lee and Lemieux [6]. Bandwidth sensitivity is assessed graphically and reported across all three thresholds.

V. Results

A. Within-Sample Performance

Table II presents the 5-fold cross-validation results. XGBoost achieves a mean accuracy of 95.5% (SD = 0.8%)

and a mean AUC of 0.984 (SD = 0.005), with precision, recall, and F1 scores uniformly above 0.94. For context, logistic regression attains only 62.9% accuracy (AUC = 0.666), random forest achieves 79.0% (AUC = 0.892), and LightGBM reaches 94.8% (AUC = 0.983), confirming that the gradient-boosted ensemble provides a substantively superior fit. Figure 1 presents the ROC curve from a representative fold (AUC = 0.989). Figure 2 shows that predicted probabilities are well-calibrated, with mean predicted probabilities closely tracking observed re-election frequencies across the full probability range. These within-sample results are impressive but, as the following subsection demonstrates, they mask a fundamental lack of temporal generalisability.

TABLE II: 5-Fold Cross-Validation Results for XGBoost (Random Split)

Fold	Accuracy	AUC	Precision	Recall	F1
1	0.9555	0.9931	0.9281	0.9810	0.9538
2	0.9496	0.9807	0.9669	0.9241	0.9450
3	0.9613	0.9790	0.9337	0.9873	0.9598
4	0.9435	0.9846	0.9481	0.9299	0.9389
5	0.9643	0.9815	0.9618	0.9618	0.9618
Mean	0.9548	0.9838	0.9477	0.9568	0.9519
SD	0.0081	0.0051	0.0162	0.0270	0.0096

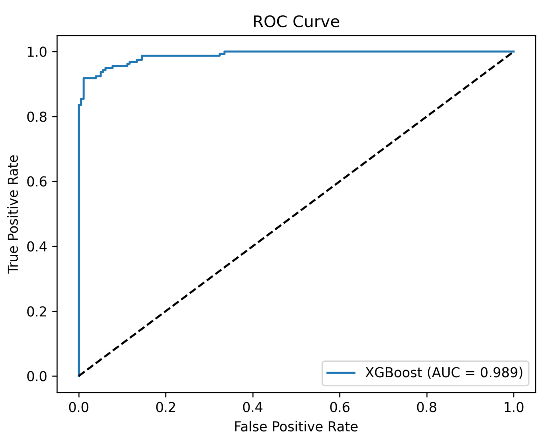

Fig. 1: ROC curve from a representative fold of the random cross-validation (AUC = 0.989), indicating strong in-sample discriminatory power.

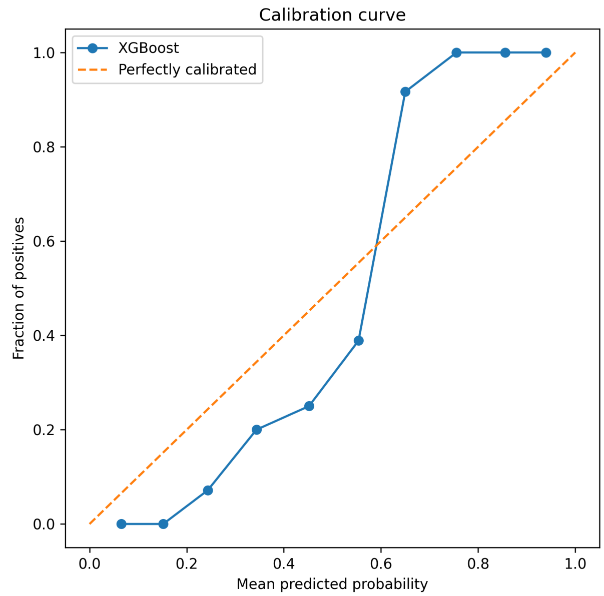

Fig. 2: Calibration curve for XGBoost predicted probabilities. Mean predicted probabilities closely track observed re-election frequencies, indicating well-calibrated predictions across the full range.

B. Temporal Generalisation Failure

Figure 3 and Table III present rolling-window test accuracies by election year. Across all eight out-of-sample test years, accuracy ranges from 37.4% to 57.3%—performance that is, at best, marginally above random guessing (50%) and in several years substantially below it. For the 2019–2022 post-Duterte period, aggregated accuracy falls to 49.4%, worse than a naïve coin-flip classifier. Crucially, this is not a pattern of gradual decay consistent with concept drift; rather, the decline is abrupt. A model trained on all elections from 1992 to 2016 fails sharply when evaluated on the 2019 and 2022 elections, consistent with a structural break at the 2016 political transition.

The election of Rodrigo Duterte to the presidency in 2016 plausibly accounts for this discontinuity. Duterte's administration further entrenched local political dynasties [8], [10] and reoriented national political discourse toward punitive populism, in which anti-crime rhetoric and extrajudicial enforcement rather than fiscal service delivery became the dominant bases of political reward [9]. Under these conditions, the fiscal–electoral relationships that characterised the 1992–2013 period ceased to be predictive.

A methodological caveat applies to the earliest test years. For the 2001 and 2004 tests, the rolling-window training sets contain only three and four election cycles, respectively (approximately 150–250 observations), which may yield unstable model estimates. However, from 2007 onward—when training sets encompass at least five cycles (≈300 observations) and grow to ten cycles (≈600 observations) by 2022—accuracy never exceeds 57.3% and averages below 50% in the post-2016 period. The temporal generalisation failure is therefore robust to the small-sample concern.

TABLE III: Rolling-Window Test Accuracies by Election Year

Test Year	Accuracy
2001	0.374
2004	0.398
2007	0.510
2010	0.441
2013	0.511
2016	0.573
2019	0.534
2022	0.494

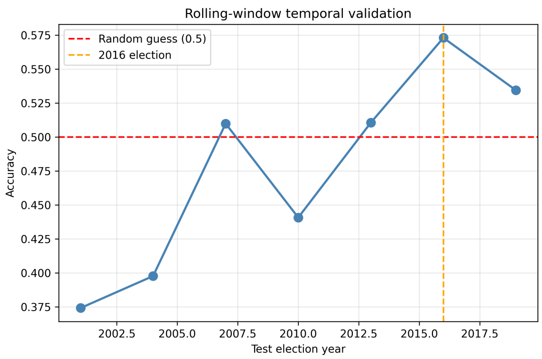

Fig. 3: Rolling-window test accuracy by election year. At each year, the model is trained on all preceding election cycles and evaluated on that year alone. The dashed reference lines mark random-guess performance (0.5) and the 2016 electoral transition. Performance is consistently near or below random, with no evidence of systematic improvement over time.

C. Feature Importance and SHAP Analysis

Table IV reports the mean absolute SHAP values from the full-data model, including interaction terms. The most influential predictor is the interaction between dynasty status and IRA dependence (`dynasty_x_ira`), closely followed by public welfare spending growth (`delta_pubwelf`) and political competition (`enc_gol`). This ordering carries a substantive interpretation: the electoral effect of fiscal dependence on national transfers is not uniform but is strongly conditioned by dynastic incumbency. The prominence of public welfare spending over health and education spending—which rank fourth and sixth, respectively—is consistent with clientelism theory rather than retrospective voting based on public goods provision.

Figures 4 and 5 present the SHAP bar plot and partial dependence plots, respectively. The partial dependence plots reveal saturation effects: marginal increases in education and health spending growth raise re-election probability up to a threshold, beyond which returns diminish. Figure 6 displays the pairwise correlation heatmap, confirming that IRA share is negatively associated with re-election while health and education spending growth are positively correlated with the outcome variable.

TABLE IV: Mean Absolute SHAP Values: Top 15 Features

Feature	Mean |SHAP|
delta_pubwelf	0.4436
dynasty_x_ira	0.4277
enc_gol	0.3748
delta_health	0.2902
ira_share	0.2776
delta_educ	0.2746
local_rev_pc	0.1876
dynasty_x_delta_pubwelf	0.1798
dynasty_x_delta_educ	0.1713
region_BARMM	0.0896
region_CARAGA	0.0646
dynasty_x_delta_health	0.0644
region_SOCCSKSARGEN	0.0643
region_CALABARZON	0.0514
prev_margin	0.0452

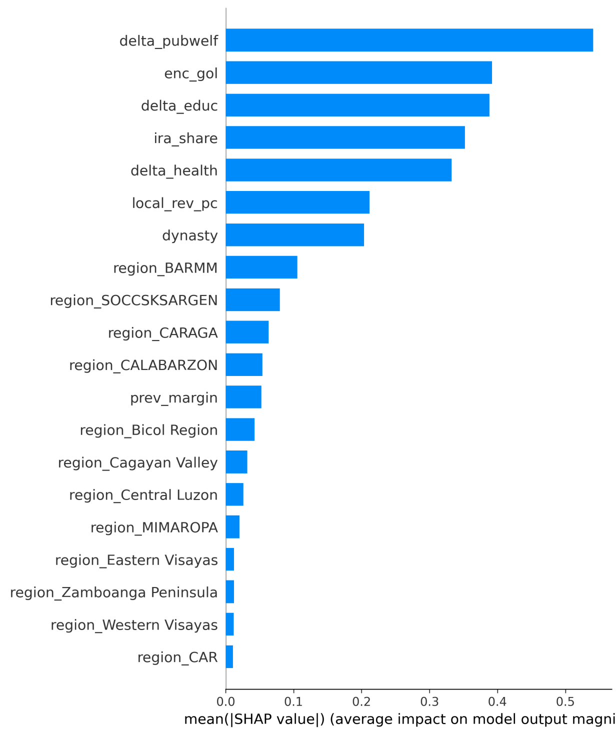

Fig. 4: Mean absolute SHAP values from the full-data model. The dynasty–IRA interaction and public welfare spending growth are the dominant predictors. Note that `delta_pubwelf` in the figure corresponds to Δpubwelf in the text.

D. Regression Discontinuity: A Virtuous Cycle

Table V presents RDD estimates for health spending growth and IRA share. At the 5% bandwidth, winning a close gubernatorial election increases subsequent health spending growth

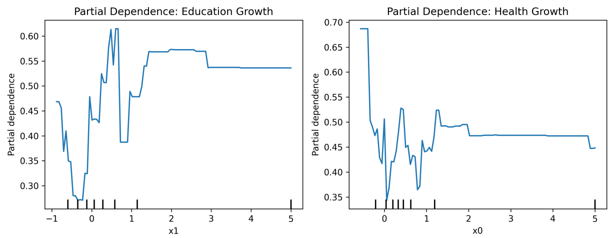

Fig. 5: Partial dependence plots for education spending growth (left) and health spending growth (right). Both relationships are positive and exhibit saturation at higher growth rates, suggesting diminishing electoral returns to fiscal effort.

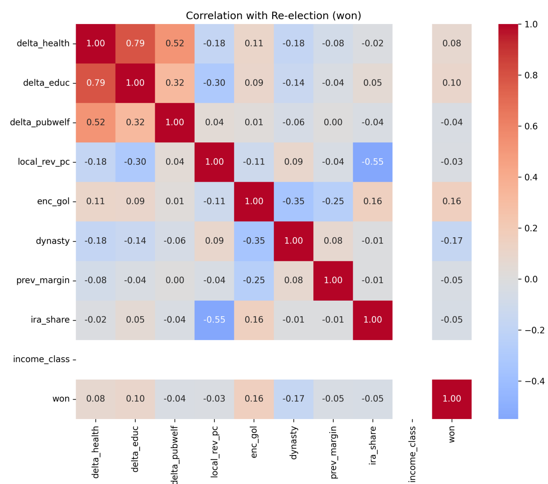

Fig. 6: Pairwise correlation heatmap. Re-election (won) is positively associated with health and education spending growth and negatively associated with IRA share, consistent with voters rewarding fiscal effort and penalising transfer dependence.

by 1.50 percentage points (SE = 0.734; 95% CI [0.07, 2.94]). The effect is positive but statistically insignificant at the narrower 3% bandwidth and the wider 7% bandwidth, reflecting sensitivity to the choice of estimation window. The effect on post-election IRA share is negative across all bandwidths but does not attain conventional significance levels. Figure 7 displays the regression discontinuity plot at the 5% bandwidth; the jump at the vote-share threshold is visually apparent. Figure 8 confirms that the estimated treatment effect remains positive across the full range of bandwidths examined.

Taken together, the predictive model and the RDD results support a *virtuous cycle* interpretation: health spending growth is positively associated with re-election in the predictive analysis, and electoral victory causally increases subsequent health investment. Neither method in isolation would reveal this dynamic; the combination of correlational and causal evidence is mutually reinforcing.

TABLE V: RDD Estimates: Health Spending Growth and IRA Share

Outcome	Bandwidth	ATE	SE	95% CI
Health growth	3%	1.300	0.912	$[-0.487, 3.087]$
Health growth	5%	1.505	0.734	$[0.066, 2.943]$
Health growth	7%	1.125	0.670	$[-0.187, 2.437]$
IRA share	3%	-0.052	0.044	$[-0.137, 0.033]$
IRA share	5%	-0.027	0.032	$[-0.089, 0.035]$
IRA share	7%	-0.012	0.029	$[-0.069, 0.044]$

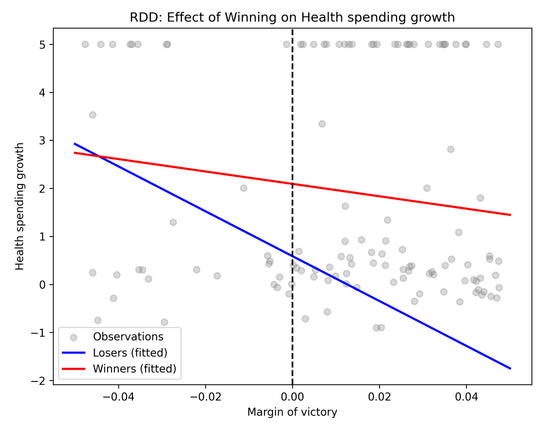

Fig. 7: Regression discontinuity plot at the 5% bandwidth. A visible jump at the vote-margin threshold (0.0) indicates a positive causal effect of electoral victory on subsequent health spending growth.

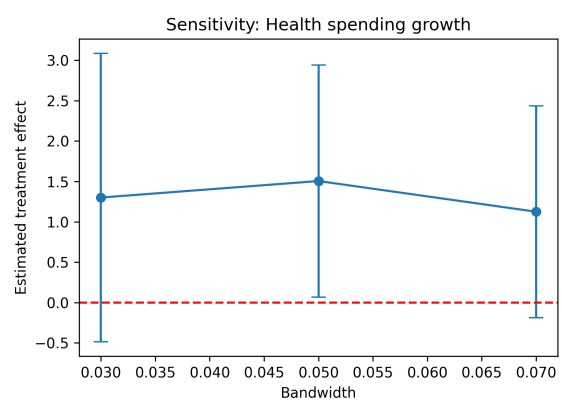

Fig. 8: Sensitivity of the RDD estimate for health spending growth to bandwidth selection. The estimated treatment effect remains positive across all bandwidths from 0.03 to 0.07, though statistical significance is achieved only at the 0.05 threshold.

E. Validity of the Regression Discontinuity Design

A necessary condition for the RDD to yield valid causal estimates is that the running variable—the incumbent's vote margin—is not subject to precise manipulation in the neighbourhood of the electoral threshold [7]. We assess this assumption using the McCrary density test, implemented via local linear regression of log-bin counts with bootstrap standard errors. The estimated log difference in densities at the 50% threshold is $\hat{\theta} = -0.0122$ (bootstrap SE = 0.0831; two-sided p-value = 0.884), indicating no statistically significant discontinuity in the density of the running variable at the cutoff. Figure 9 presents the density histogram; the distribution is smooth and continuous across the threshold. The x-axis spans -0.25 to 0.5 because the data contain no vote margins below -0.25, a truncation that does not affect the validity of the test near the cutoff. These results provide no evidence of electoral manipulation and support the internal validity of the RDD estimates.

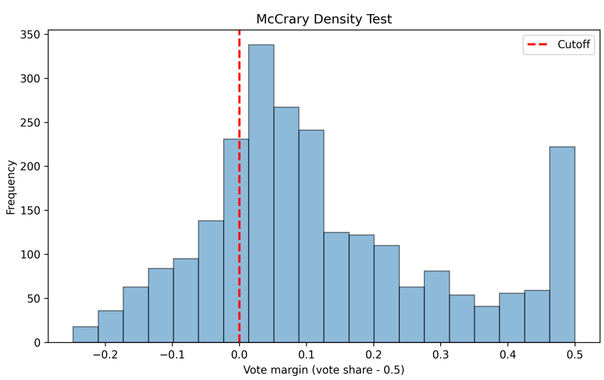

Fig. 9: McCrary density test of the incumbent vote margin. The vertical dashed line marks the 50% electoral threshold. The distribution is smooth and continuous at the cutoff ($\hat{\theta} = -0.0122$, $p = 0.884$), consistent with the absence of systematic vote-margin manipulation. The x-axis spans -0.25 to 0.5 because no observations fall below -0.25.

VI. Discussion

A. Temporal Instability as the Central Finding

The most consequential result of this study is the temporal generalisation failure. The model's high within-sample cross-validation accuracy (95.5%) is an artefact of the random-split evaluation protocol; it does not reflect a stable, generalisable fiscal–electoral relationship. When subjected to temporally honest evaluation—training on historical data and predicting future elections—the model performs at or below chance. This finding has a direct implication for the methodology of electoral accountability research: *predictive models evaluated solely on randomly split hold-out sets may substantially overstate the stability and generalisability of the relationships they identify.* Temporal validation is not a supplementary robustness check; it is a minimum standard of methodological rigour for research using non-stationary political panel data.

The sharp deterioration in predictive performance after 2016 is consistent with a regime shift in Philippine local politics. The Duterte administration's populist rhetoric, centralisation of discretionary transfers, and tolerance of local dynastic networks appear to have reconfigured the basis of electoral reward at the subnational level [9], [10]. Fiscal accountability—to the extent it existed prior to 2016—may have been supplanted by partisan loyalty, personalised leadership appeals, and clientelistic payoffs that are not captured by the fiscal variables available in the dataset.

B. Synthesising ML and RDD: A Virtuous Cycle

Read in conjunction, the predictive model and the RDD results yield a coherent and theoretically interpretable narrative. The predictive model establishes that health spending growth is among the strongest fiscal correlates of gubernatorial re-election in the historical data. The RDD establishes that narrowly winning a gubernatorial election causally increases subsequent health investment. These two findings are mutually consistent: incumbents who invest in health are more likely to survive close elections (retrospective accountability), and incumbents who survive close elections are more likely to continue health investment (incumbency advantage). Neither the correlational nor the causal analysis in isolation would reveal this self-reinforcing dynamic; the combination of methods is essential.

The fragility of the RDD effect—statistically significant only at the 5% bandwidth—warrants interpretive caution. The directional consistency of the estimates across all three bandwidths, however, provides some assurance that the null result at the 3% and 7% thresholds reflects sampling variance rather than the absence of an effect.

C. Policy Implications

The policy implications of these findings are substantively important but cautionary in tenor. Fiscal performance—particularly in health and public welfare—was associated with electoral reward in the Philippine provinces during the 1992–2013 period, suggesting that accountability mechanisms were at least partially operative. However, this relationship appears to have broken down in the post-2016 period. Accountability mechanisms are evidently susceptible to disruption by national political shocks, and policies aimed at strengthening local fiscal transparency or rewarding fiscal performance may have limited electoral traction if voters' priorities have shifted toward non-fiscal dimensions of governance.

VII. Limitations

Several limitations constrain the scope of inference. First, the predictive model is correlational rather than causal; observed associations between fiscal variables and re-election may reflect common causes rather than an accountability mechanism. Second, temporal generalisation fails after 2016, limiting the model's predictive utility for contemporary elections. Third, the RDD effect is bandwidth-sensitive and achieves conventional significance only at a single threshold, tempering causal claims. Fourth, the dataset is restricted to provincial governors; the electoral accountability dynamics of municipalities and cities, which face different fiscal and political constraints, may differ substantially. Fifth, the sample is confined to a single income class, precluding cross-class heterogeneity analysis. Finally, the 2016 structural break cannot be cleanly attributed to the Duterte transition alone; other contemporaneous changes—including infrastructure programme expansion and rising conditional cash transfer coverage—may independently account for shifts in voter behaviour.

VIII. Conclusion

This study provides the first application of rolling-window temporal validation to a fiscal–electoral accountability model in the Philippines, and in doing so reveals that high within-sample accuracy is insufficient evidence of a stable political relationship. Using a 30-year panel of Philippine provinces, an XGBoost classifier, SHAP-based feature attribution, and a regression discontinuity design, we show that fiscal performance predicted gubernatorial re-election during the 1992–2016 period but ceased to do so thereafter. The dynasty–IRA interaction is the dominant predictor, consistent with a clientelistic rather than programmatic accountability mechanism. A fragile but directionally consistent RDD result suggests that electoral victory causes increased health investment, supporting a virtuous cycle interpretation when combined with the predictive findings. The post-2016 collapse in predictive performance constitutes evidence of a regime shift in Philippine subnational politics, with direct implications for the design of local accountability institutions under Sustainable Development Goal 16. Future research should prioritise temporal validation as a methodological standard, investigate the mechanisms underlying the 2016 regime shift, and extend the analysis to municipal-level governance.

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Use of Generative AI

Generative AI tools were used for grammar checking, style editing, and structural suggestions in the discussion section. All AI-assisted content was independently reviewed and revised by the author to ensure accuracy and coherence with the research findings.

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